國立陽明交通大學
NATIONAL YANG MING CHIAO TUNG UNIVERSITY

Institute of Artificial Intelligence Innovation
Department of Computer Science

Embedded Operating System

HW03: Add Modules into Kernel

Shuo-Han Chen (陳碩漢)
shch@nycu.edu.tw

Wed. 13:20 – 16:20 EDB06

Purpose for this Homework

- **Part 1. Learn how to compile Linux Kernel**
  - You need to download the kernel source code that matches your current kernel version, compile it, and then install it.
- **Part 2. Learn add your own module into Linux Kernel & Install Kernel**
  - After learning how to compile the Linux kernel yourself
  - You should proceed by incorporating your previously created 'Hello World' module into the kernel source code.
  - Subsequently, compile the kernel with the newly added module and proceed to run it.

Part 1. Learn how to compile Linux Kernel

1. Check your kernel version by running
   1. cat /proc/version_signature

```
liaw@liaw:~/Desktop$ cat /proc/version_signature
Ubuntu 6.5.0-27.28~22.04.1-generic 6.5.13
```

This is your kernel version

2. Download the Linux kernel source code
   1. wget https://cdn.kernel.org/pub/linux/kernel/v6.x/linux-6.5.13.tar.xz
   2. tar Jxvf linux-6.5.13.tar.xz
   3. cd linux-6.5.13

Adjust those two place to match your version

3. Compile Step
   1. Download dependencies by running
      1. sudo apt-get install build-essential libncurses-dev bison flex libssl-dev libelf-dev dwarves
   2. Create .config by running
      1. make menuconfig

      1. You should see a configuration page
      2. Select `save` and then `exit`

   3. Modify some `.config` value
      1. sudo vim .config

      1. CONFIG_SYSTEM_TRUSTED_KEYS = ""
      2. CONFIG_SYSTEM_REVOCATION_KEYS = ""

   4. Make
      1. make –j 8

      nproc

      1. During the process, you might experience interruptions where you need to enter 'make' multiple times to complete the compilation.
      2. Once finished, you'll see the message 'bzImage is already done'.

4. Install kernel
   1. sudo make INSTALL_MOD_STRIP=1 modules_install
   2. sudo make INSTALL_MOD_STRIP=1 install
   3. sudo make headers_install INSTALL_HDR_PATH=/usr

```
  BTF [M] net/nsh/nsh.ko
  BTF [M] net/hsr/hsr.ko
  BTF [M] net/qrtr/qrtr.ko
  BTF [M] net/qrtr/qrtr-smd.ko
  BTF [M] net/qrtr/qrtr-tun.ko
  BTF [M] net/qrtr/qrtr-mhi.ko
  BTF [M] virt/lib/irqbypass.ko
liaw@liaw:~/linux-6.5.13$ make -j 8
  DESCEND objtool
  DESCEND bpf/resolve_btfids
  INSTALL libsubcmd_headers
  INSTALL libsubcmd_headers
  CALL    scripts/checksyscalls.sh
  CHK     kernel/kheaders_data.tar.xz
Kernel: arch/x86/boot/bzImage is ready  (#2)
liaw@liaw:~/linux-6.5.13$
```

Part 2. Learn add your own module into Linux Kernel & Install Kernel

1. Add your module to linux kernel source
    1. Find the hello world kernel module you wrote in HW01
    2. Add your student ID to the print message -> “{StudentID}: Hello World !\n”
    3. Follow steps here: https://kerneltweaks.wordpress.com/2013/11/26/add-new-module-to-linux-kernel-source-code/
        - If your platform is X86, skip the ARM related part.
2. Find your print message in dmesg after your compiled kernel is installed.

```
[    2.734811] Asymmetric key parser 'x509' registered
[    2.734918] Block layer SCSI generic (bsg) driver version 0.4 loaded (major 243)
[    2.735146] io scheduler mq-deadline registered
[    2.736135] 312581029: Hello world !
[    2.736778] pcieport 0000:00:15.0: PME: Signaling with IRQ 24
[    2.736837] pcieport 0000:00:15.0: pciehp: Slot #160 AttnBtn+ PwrCtrl+ MRL- AttnI
```

Requirement of this Homework

1. Screenshot the console message of your compiled Linux Kernel
2. Screenshot the Hello World message from dmesg after your new Linux Kernel is installed

Submission

- Submit your PDF report in a zip format to the google Form.
- The google form will be published in Teams before the Deadline.
- The zip folder structure should look like this:
  - HW03_studentID.zip
    - report.pdf

Not like this:

- HW03_studentID.zip
  - folder
    - report.pdf

Grading

- Part 1. Learn how to compile Linux Kernel - 30%
- Part 2. Learn add your own module into Linux Kernel & Install Kernel - 70%
- If your report format or submission folder structure does not meet the requirements, 20% will be deducted from your total score. Please double-check before submitting.

Q & A

Remember to submit your report.

The deadline is set as 5/21 23:59

Late submission will be not accepted!